# Vector Spaces and Subspaces Reading



In this lesson we describe how to expand the concept of Euclidean space and subspaces from $\mathbb{R}^n$ to a more general setting. To get us started, let $\mathbf{P}^2$ be the set of all polynomials with real coefficients that have degree 2 or less. A typical element of $\mathbf{P}^2$ has the form $p(x) = a_2x^2 + a_1x + a_0$, where $a_2$, $a_1$, and $a_0$ are real numbers. Let's compare $\mathbf{R}^3$ and $\mathbf{P}^2$.

- If $p(x) = a_2x^2 + a_1x + a_0$ and $q(x) = b_2x^2 + b_1x + b_0$ are two polynomials in $\mathbf{P}^2$, then the sum is $p(x) + q(x) = (a_2 + b_2)x^2 + (a_1 + b_1)x + (a_0 + b_0)$

  When adding polynomials, we add together the coefficients of like terms. This is similar to the componentwise addition of elements in $\mathbf{R}^3$.
- If $c$ is a real number and $p(x)$ is as above, then $cp(x) = (ca_2)x^2 + (ca_1)x + (ca_0)$

  Scalar multiplication of polynomials distributes across terms, similar to how scalar multiples distribute across components in $\mathbf{R}^3$.
- Just as $\mathbf{R}^3$ is closed under addition and scalar multiplication, so is $\mathbf{P}^2$. The sum of two polynomials in $\mathbf{P}^2$ has degree no greater than 2, as is the degree of the scalar multiple of a polynomial in $\mathbf{P}^2$. (Note that these operations might decrease the degree but cannot increase it.)
- The zero polynomial $z(x) = 0$ satisfies $p(x) + z(x) = p(x)$ for every polynomial in $\mathbf{P}^2$, so that $z(x)$ plays the same role as $\mathbf{0} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$ does in $\mathbf{R}^3$.
- For every polynomial $p(x)$ in $\mathbf{P}^2$, there is another polynomial $-p(x) = -a_2x^2 - a_1x - a_0$ such that $p(x) + (-p(x)) = 0 = z(x)$. This is also true in $\mathbf{R}^3$, where for each $\begin{bmatrix} a \\ b \\ c \end{bmatrix}$ there exists $\begin{bmatrix} -a \\ -b \\ -c \end{bmatrix}$ such that $\begin{bmatrix} a \\ b \\ c \end{bmatrix} + \begin{bmatrix} -a \\ -b \\ -c \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} = \mathbf{0}$.
- The definitions of addition and scalar multiplication on $\mathbf{P}^2$ satisfy the same distributive and associative laws as those of $\mathbf{R}^3$.

Although they look different, $\mathbf{R}^3$ and $\mathbf{P}^2$ have many similar features. The preceding discussion shows that the polynomials in $\mathbf{P}^2$ have similar properties. Other sets of mathematical objects, such as matrices and continuous functions, also possess these properties.

**DEFINITION 1**

A vector space consists of a set $V$ of vectors together with operations of addition and scalar multiplication on the vectors that satisfy each of the following:

1. If $\mathbf{v}_1$ and $\mathbf{v}_2$ are in $V$, then so is $\mathbf{v}_1 + \mathbf{v}_2$. ($V$ is closed under addition.)
2. If $c$ is a real scalar and $\mathbf{v}$ is in $V$, then so is $c\mathbf{v}$. ($V$ is closed under scalar multiplication.)
3. There exists a **zero vector** $\mathbf{0}$ in $V$ such that $\mathbf{0} + \mathbf{v} = \mathbf{v}$ for all $\mathbf{v}$ in $V$.
4. For each $\mathbf{v}$ in $V$, there exists an **additive inverse** (or **opposite**) vector $-\mathbf{v}$ in $V$ such that $\mathbf{v} + (-\mathbf{v}) = \mathbf{0}$ for all $\mathbf{v}$ in $V$.
5. For all $\mathbf{v}_1$, $\mathbf{v}_2$, and $\mathbf{v}_3$ in $V$ and real scalars $c_1$ and $c_2$, we have
   a. $\mathbf{v}_1 + \mathbf{v}_2 = \mathbf{v}_2 + \mathbf{v}_1$
   b. $(\mathbf{v}_1 + \mathbf{v}_2) + \mathbf{v}_3 = \mathbf{v}_1 + (\mathbf{v}_2 + \mathbf{v}_3)$
   c. $c_1(\mathbf{v}_1 + \mathbf{v}_2) = c_1\mathbf{v}_1 + c_1\mathbf{v}_2$
   d. $(c_1 + c_2)\mathbf{v}_1 = c_1\mathbf{v}_1 + c_2\mathbf{v}_1$
   e. $(c_1c_2)\mathbf{v}_1 = c_1(c_2\mathbf{v}_1)$
   f. $1 \cdot \mathbf{v}_1 = \mathbf{v}_1$

Definition 1 is technically the definition of a *real* vector space, and we could replace the real scalars with complex numbers with minimal changes. But for now real scalars will suffice, so from here on we assume that "vector space" refers to a real vector space unless stated otherwise.

Note that Euclidean space $\mathbf{R}^n$ is a vector space. $\mathbf{P}^2$, together with the usual operations of addition and scalar multiplication, also forms a vector space.

Three important points:

- To describe a vector space, we need to specify both the set of vectors and the arithmetic operations (addition and scalar multiplication) that are performed on them. The set alone is not enough.
- **Vectors are not always columns of numbers!** (For instance, in $\mathbf{P}^2$ the polynomials are the vectors.) This takes getting used to but is crucial, so say it to yourself every night as you fall asleep until it sinks in. It is fine to think of a column of numbers as an example of a vector, as long as you do not assume that a vector is always a column of numbers.
- The phrase "vector space" refers to any set satisfying Definition 1. The phrase "Euclidean space" will be used for the specific vector space $\mathbf{R}^n$ with the standard definition of addition and scalar multiplication.

**Example 1**

Let $V = \mathbb{R}^{2\times2}$ denote the set of real $2 \times 2$ matrices, together with the usual definition of matrix addition and multiplication by a constant scalar. Show that $\mathbb{R}^{2\times2}$ is a vector space.

**Solution**

We have a clearly defined set of vectors (the real $2 \times 2$ matrices) and definitions for addition and scalar multiplication. It remains to verify that the five conditions of Definition 1 hold:

1. If $A$ and $B$ are real $2 \times 2$ matrices, then we already know that $A + B$ is also a real $2 \times 2$ matrix. Hence $\mathbb{R}^{2\times2}$ is closed under addition.
2. If $c$ is a real scalar and $A$ is in $\mathbb{R}^{2\times2}$, then $cA$ is also a real $2 \times 2$ matrix. Thus $\mathbb{R}^{2\times2}$ is closed under scalar multiplication.
3. If $0_{22} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$ and $A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$ then $0_{22} + A = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} + \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} = \begin{bmatrix} (0 + a_{11}) & (0 + a_{12}) \\ (0 + a_{21}) & (0 + a_{22}) \end{bmatrix} = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} = A$

   Hence $0_{22} + A = A$ for all real $2 \times 2$ matrices, so that $0_{22}$ is the zero vector in $\mathbb{R}^{2\times2}$.
4. If $A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$ then $-A = \begin{bmatrix} -a_{11} & -a_{12} \\ -a_{21} & -a_{22} \end{bmatrix}$ satisfies $A + (-A) = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} + \begin{bmatrix} -a_{11} & -a_{12} \\ -a_{21} & -a_{22} \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} = 0_{22}$

   so that each vector in $\mathbb{R}^{2\times2}$ has an additive inverse.
5. The six conditions (a)–(f) all follow directly from properties of the real numbers. Verification is left as an exercise.

Since all the required conditions hold, the set $\mathbb{R}^{2\times2}$ together with the given arithmetic operations form a vector space.

**Example 2**

Let $\mathbf{Q}^2$ denote the set of polynomials with real coefficients that have degree equal to 2, together with the usual definition of addition and scalar multiplication for polynomials. Is $\mathbf{Q}^2$ a vector space?

**Solution**

The set $\mathbf{Q}^2$ satisfies some of the conditions of Definition 1, but it falls short on others. For instance, if $q_1(x) = x^2$ and $q_2(x) = 5 - x^2$, then $q_1$ and $q_2$ are both in $\mathbf{Q}^2$, but $(q_1 + q_2)(x) = q_1(x) + q_2(x) = 5$ is not, so $\mathbf{Q}^2$ is not closed under addition. Hence $\mathbf{Q}^2$ is not a vector space.

$\mathbf{Q}^2$ consists of polynomials of degree 2, whereas $\mathbf{P}^2$ consists of the polynomials of degree 2 or less.

Before moving on to the next example, we pause to report several properties of vector spaces that are consequences of Definition 1.

**THEOREM 2**

Let $V$ be a vector space and suppose that $\mathbf{v}$ is in $V$. Then:

(a) $0\mathbf{v}$ is a zero vector of $V$, then $\mathbf{v} + 0 = \mathbf{v}$.

(b) If $-\mathbf{v}$ is an additive inverse of $\mathbf{v}$, then $-\mathbf{v} + \mathbf{v} = \mathbf{0}$.

(c) $\mathbf{v}$ has a unique additive inverse $-\mathbf{v}$.

(d) The zero vector $\mathbf{0}$ is unique.

(e) $0 \cdot \mathbf{v} = \mathbf{0}$.

(f) $(-1)\cdot\mathbf{v} = -\mathbf{v}$.

**Proof**

These may seem obvious based upon past experience with real numbers, but remember, all we can assume about a vector space are the properties given in Definition 1. We give a proof of (a) and (c), and leave the rest as an exercise.

For (a), we know from (3) of Definition 1 that there exists a zero vector $\mathbf{0}$ such that

$$0 + \mathbf{v} = \mathbf{v}$$

Property (5a) of Definition 1 (the Commutative Law) lets us interchange the order of addition. Doing so on the left above gives us the equation

$$\mathbf{v} + 0 = \mathbf{v}$$

proving part (a) of our theorem.

For part (c), note that (4) of Definition 1 states that every vector in $V$ has at least one additive inverse. To prove that the additive inverse is unique, we suppose to the contrary that there exists a vector $\mathbf{v}$ in $V$ with two additive inverses $\mathbf{v}_1$ and $\mathbf{v}_2$. Then

$$\mathbf{v} + \mathbf{v}_1 = 0 \text{ and } \mathbf{v} + \mathbf{v}_2 = 0$$

so that

$$\mathbf{v} + \mathbf{v}_1 = \mathbf{v} + \mathbf{v}_2$$

Property (5a) of Definition 1 (the Commutative Law) lets us interchange the order of addition, so we also have

$$\mathbf{v}_1 + \mathbf{v} = \mathbf{v}_2 + \mathbf{v}$$

To cancel out $\mathbf{v}$, we start by adding $\mathbf{v}_1$ to both sides of the equation.

$$(\mathbf{v}_1 + \mathbf{v}) + \mathbf{v}_1 = (\mathbf{v}_2 + \mathbf{v}) + \mathbf{v}_1 \Rightarrow \mathbf{v}_1 + (\mathbf{v} + \mathbf{v}_1) = \mathbf{v}_2 + (\mathbf{v} + \mathbf{v}_1)$$

The grouping on the right is justified by (5b) of Definition 1 (the Associative Law). Since $\mathbf{v} + \mathbf{v}_1 = \mathbf{v} + \mathbf{v}_2 = 0$, our equation simplifies to

$$\mathbf{v}_1 + 0 = \mathbf{v}_2 + 0 \Rightarrow \mathbf{v}_1 = \mathbf{v}_2$$

with the cancellation of the zero vector 0 justified by part (a) of this theorem. Since $\mathbf{v}_1 = \mathbf{v}_2$, it follows that the additive inverse is unique.

**Example 3**

Let $\mathbb{R}^\infty$ denote the set of all infinite sequences of real numbers $\mathbf{v} = (v_1, v_2, \ldots)$, so the elements in $\mathbb{R}^\infty$ have an infinite number of components. Addition and scalar multiplication are defined componentwise, by

$$(v_1, v_2, \ldots) + (u_1, u_2, \ldots) = (v_1 + u_1, v_2 + u_2, \ldots)$$

and

$$c(v_1, v_2, \ldots) = (cv_1, cv_2, \ldots)$$

Show that $\mathbb{R}^\infty$ is a vector space.

**Solution**

Here $\mathbb{R}^\infty$ looks somewhat like $\mathbf{R}^n$ but with an infinite number of components. It is not hard to see that $\mathbb{R}^\infty$ is closed under addition and scalar multiplication, and that the zero vector is given by

$$0 = (0, 0, 0, \ldots)$$

Also, if $\mathbf{v} = (v_1, v_2, \ldots)$, then $-\mathbf{v} = (-v_1, -v_2, \ldots)$ satisfies $\mathbf{v} + (-\mathbf{v}) = 0$, so each vector in $\mathbb{R}^\infty$ has an additive inverse. Finally, the six conditions given in (5) of Definition 1 are inherited from the real numbers. Hence $\mathbb{R}^\infty$ is a vector space.

**Example 4**

Determine if $V$ is a vector space, where $V$ is the set of vectors of the form $\begin{bmatrix} a \\ b \end{bmatrix}$ (*a* and *b* real numbers) together with addition and scalar multiplication defined by

$$\begin{bmatrix} a_1 \\ b_1 \end{bmatrix} \oplus \begin{bmatrix} a_2 \\ b_2 \end{bmatrix} = \begin{bmatrix} a_1 + a_2 \\ b_1 + b_2 \end{bmatrix} \text{ and } c \odot \begin{bmatrix} a \\ b \end{bmatrix} = \begin{bmatrix} ca \\ 0 \end{bmatrix}$$

**Solution**

We answer this question by working our way through the conditions of Definition 1.

1. Addition is defined as in Euclidean space, so $V$ is closed under addition.
2. Scalar multiplication is different from Euclidean space, but $c\mathbf{v}$ is in $V$ for any *c* and $\mathbf{v}$ so $V$ is closed under scalar multiplication.
3. The zero vector $\mathbf{0} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$ for Euclidean space is also the zero vector of $V$.
4. Addition is defined as in Euclidean space, so for every $\mathbf{v}$ in $V$ there is also an additive inverse $-\mathbf{v}$ in $V$.
5. The commutative and associative laws of addition are inherited from Euclidean space, so (a) and (b) are satisfied. The other parts of (5) involve scalar multiplication, so they should be verified more carefully. We let $\mathbf{v}_1 = \begin{bmatrix} a_1 \\ b_1 \end{bmatrix}$ and $\mathbf{v}_2 = \begin{bmatrix} a_2 \\ b_2 \end{bmatrix}$

   (c) $c_1(\mathbf{v}_1 + \mathbf{v}_2) = c_1\left(\begin{bmatrix} a_1 \\ b_1 \end{bmatrix} + \begin{bmatrix} a_2 \\ b_2 \end{bmatrix}\right) = c_1\begin{bmatrix} a_1 + a_2 \\ b_1 + b_2 \end{bmatrix} = \begin{bmatrix} c_1(a_1 + a_2) \\ 0 \end{bmatrix}$
   $= \begin{bmatrix} c_1a_1 + c_1a_2 \\ 0 \end{bmatrix} = \begin{bmatrix} c_1a_1 \\ 0 \end{bmatrix} \oplus \begin{bmatrix} c_1a_2 \\ 0 \end{bmatrix} = c_1\begin{bmatrix} a_1 \\ b_1 \end{bmatrix} \oplus c_1\begin{bmatrix} a_2 \\ b_2 \end{bmatrix}$
   $= c_1\mathbf{v}_1 \oplus c_1\mathbf{v}_2$

   (d) $(c_1 + c_2)\mathbf{v}_1 = (c_1 + c_2)\begin{bmatrix} a_1 \\ b_1 \end{bmatrix} = \begin{bmatrix} (c_1 + c_2)a_1 \\ 0 \end{bmatrix} = \begin{bmatrix} c_1a_1 + c_2a_1 \\ 0 \end{bmatrix}$
   $= \begin{bmatrix} c_1a_1 \\ 0 \end{bmatrix} \oplus \begin{bmatrix} c_2a_1 \\ 0 \end{bmatrix} = c_1\begin{bmatrix} a_1 \\ b_1 \end{bmatrix} \oplus c_2\begin{bmatrix} a_1 \\ b_1 \end{bmatrix} = c_1\mathbf{v}_1 \oplus c_2\mathbf{v}_1$

   (e) $(c_1c_2)\mathbf{v}_1 = (c_1c_2)\begin{bmatrix} a_1 \\ b_1 \end{bmatrix} = \begin{bmatrix} (c_1c_2)a_1 \\ 0 \end{bmatrix} = \begin{bmatrix} c_1(c_2a_1) \\ 0 \end{bmatrix}$
   $= c_1\left(\begin{bmatrix} c_2a_1 \\ 0 \end{bmatrix}\right) = c_1\left(c_2\begin{bmatrix} a_1 \\ b_1 \end{bmatrix}\right) = c_1(c_2\mathbf{v}_1)$

Thus for every required condition has been satisfied. However, condition (f) is not because

$$1 \cdot \mathbf{v}_1 = 1 \cdot \begin{bmatrix} a_1 \\ b_1 \end{bmatrix} = \begin{bmatrix} a_1 \\ 0 \end{bmatrix} \neq \mathbf{v}_1$$

when $b_1 \neq 0$. Thus $V$ is not quite a vector space.

**Example 5**

Suppose that $n$ < $n$ are real numbers, and let $C[a, b]$ denote the set of all real-valued continuous functions on $[a, b]$. For $f$ and $g$ in $C[a, b]$ and a real scalar $c$, we define $(f + g)(x) = f(x) + g(x)$ and $(cf)(x) = c \cdot f(x)$. (These are the usual pointwise definitions of addition and scalar multiplication of functions.) Show that $C[a, b]$ is a vector space.

**Solution**

The set $C[a, b]$ looks different from the other vector spaces we have seen, where the vectors had entries reminiscent of components of vectors in $\mathbf{R}^n$. Here, we have a set consisting of functions such as $f(x) = \sin(x)$ and $g(x) = e^{-x}$.

To determine if $C[a, b]$ is a vector space, we avoid being distracted by superficial appearances by focusing on the definition. First, the set $C[a, b]$ together with addition and scalar multiplication are clearly defined. Next we need to determine if the five conditions of Definition 1 are all met. We verify the first three here and leave the remaining conditions as an exercise.

1. Is $C[a, b]$ closed under addition?

   If $f$ and $g$ are both in $C[a, b]$, then both are continuous on the interval $[a, b]$. The sum of two continuous functions is also a continuous function, so that $f + g$ is in $C[a, b]$. Hence $C[a, b]$ is closed under addition.
2. Is $C[a, b]$ closed under scalar multiplication?

   If $f$ is in $C[a, b]$ and $c$ is a scalar, then $c \cdot f(x)$ is continuous on $[a, b]$, so that $cf$ is also in $C[a, b]$. Thus $C[a, b]$ is also closed under scalar multiplication.
3. Is there a zero function?

   If $z(x) = 0$, the identically zero function, then $z(x)$ is continuous on $[a, b]$ and so is in $C[a, b]$. Thus if $f$ is in $C[a, b]$, then $z(x) + f(x) = f(x)$ for all $x$ in $[a, b]$. Hence $z + f = f$ for all $f$ in $C[a, b]$, so that $z$ is the zero function.

The remaining questions that need to be answered are:

4. Does each function have an additive inverse?
5. Do the six commutative, distributive, and associative laws hold?

Verifying that the answer to each is yes is left as an exercise. Therefore $C[a, b]$ is a vector space.

Below is a brief list of vector spaces. Some have already been verified as vector spaces, and others are left as exercises.

- Euclidean space $\mathbf{R}^n$ in $>$ 0 an integer, together with the standard addition and scalar multiplication of vectors.
- $\mathbf{P}^n$ (*n* ≥ 1 an integer, the set of polynomials with real coefficients and degree no greater then *n*, together with the usual addition and scalar multiplication of polynomials.
- $\mathbb{R}^{m\times n}$, the set of real $m \times n$ matrices together with the usual definition of matrix addition and scalar multiplication.
- $\mathbf{P}$, the set of polynomials with real coefficients and any degree, together with the usual addition and scalar multiplication of polynomials.
- $\mathbb{R}^\infty$, together with addition and scalar multiplication of vectors described in Example 3.
- $C[a, b]$, the set of real-valued continuous functions on the interval $[a, b]$, together with the usual addition and scalar multiplication of functions.
- $C(\mathbb{R})$, the set of real-valued continuous functions on the real numbers $\mathbb{R}$, together with the usual addition and scalar multiplication of functions.
- $T(m, n)$, the set of linear transformations $T : \mathbb{R}^m \to \mathbb{R}^n$, together with the usual addition and scalar multiplication of functions.

This list is far from exhaustive. There are many other vector spaces not included.

## Subspaces

Just as in Euclidean space, a subspace of a vector space can be thought of as a vector space contained within another vector space. Since a subspace inherits the properties of the parent vector space, all that is required to certify its subspace status is that it be closed under addition and scalar multiplication. The formal definition is essentially identical to the one given for Euclidean space.

**DEFINITION 3**

A subset $S$ of a vector space $V$ is a **subspace** if $S$ satisfies the following three conditions:

(a) $S$ contains $\mathbf{0}$, the zero vector.

(b) If $\mathbf{u}$ and $\mathbf{v}$ are in $S$, then $\mathbf{u} + \mathbf{v}$ is also in $S$.

(c) If $c$ is a scalar and $\mathbf{v}$ is in $S$, then $c\mathbf{v}$ is also in $S$.

**Example 6**

Let $S$ denote the set of all polynomials $p$ with real coefficients such that $p(0) = 0$. Is $S$ a subspace of $\mathbf{P}$, the set of all polynomials with real coefficients?

**Solution**

First, we note that $S$ is a subset of $\mathbf{P}$. Next, we do the same thing that we did to show that a subset of $\mathbf{R}^n$ is a subspace: We check if the three conditions of the definition hold.

1. The zero polynomial $z(x) = 0$ (the identically zero function) satisfies $z(0) = 0$ and hence is in $S$, so $S$ contains the zero vector.
2. If $p$ and $q$ are both in $S$, then $p(0) = 0$ and $q(0) = 0$. Therefore $(p + q)(0) = p(0) + q(0) = 0$, so that $p + q$ is also in $S$.
3. If $p$ is in $S$ and $c$ is a real scalar, then $(cp)(0) = c(p(0)) = 0$. Hence $cp$ is also in $S$.

Since the three conditions of Definition 3 hold, $S$ is a subspace of $\mathbf{P}$.

**Example 7**

Suppose that $m$ < $n$ are both integers. Is $\mathbf{P}^m$ a subspace of $\mathbf{P}^n$?

**Solution**

First, we note that $\mathbf{P}^m$ is a subset of $\mathbf{P}^n$. Moreover, since $\mathbf{P}^m$ is itself a vector space, we know that it contains the zero vector, is closed under addition, and is closed under scalar multiplication. Thus the three subspace conditions are met, so that $\mathbf{P}^m$ is a subspace of $\mathbf{P}^n$.

**Example 8**

Suppose that $m$ < $n$ are both integers. Is $\mathbb{R}^m$ a subspace of $\mathbf{R}^n$?

**Solution**

While it is true that $\mathbb{R}^m$ is itself a vector space, it is not a subset of $\mathbf{R}^n$ (the vectors in $\mathbb{R}^m$ have fewer components than those in $\mathbf{R}^n$) and so cannot be a subspace of $\mathbf{R}^n$.

**Example 9**

The **trace** of a square matrix is the sum of the diagonal terms. Suppose that $S$ is the subset of $\mathbb{R}^{2\times2}$ consisting of matrices with trace equal to 0. Is $S$ a subspace of $\mathbb{R}^{2\times2}$?

**Solution**

By definition $S$ is a subset of $\mathbb{R}^{2\times2}$. Let's check the three conditions of Definition 3.

1. The zero matrix of $\mathbb{R}^{2\times2}$ is $0_{22} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$ which has trace 0. Hence $0_{22}$ is in $S$.
2. Suppose that $A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$ and $B = \begin{bmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{bmatrix}$ are both in $S$. Then $a_{11} + a_{22} = 0$ and $b_{11} + b_{22} = 0$. Since $A + B = \begin{bmatrix} (a_{11} + b_{11}) & (a_{12} + b_{12}) \\ (a_{21} + b_{21}) & (a_{22} + b_{22}) \end{bmatrix}$ the trace of $A + B$ is equal to $(a_{11} + b_{11}) + (a_{22} + b_{22}) = (a_{11} + a_{22}) + (b_{11} + b_{22}) = 0$ Hence $A + B$ is also in $S$.
3. For a scalar $c$ and $A$ in $S$ as in (b), we have $cA = \begin{bmatrix} ca_{11} & ca_{12} \\ ca_{21} & ca_{22} \end{bmatrix}$ Therefore the trace of $cA$ is $ca_{11} + ca_{22} = c(a_{11} + a_{22}) = 0$ which implies that $cA$ is also in $S$.

Since all three conditions for a subspace are satisfied, $S$ is a subspace.

The trivial subspaces carry over from Euclidean space.

**THEOREM 4**

Suppose that $V$ is a vector space. Then $S = \{0\}$ and $S = V$ are both subspaces of $V$, sometimes called the trivial subspaces.

The proof is left as an exercise. The last two examples require some knowledge of calculus.

**Example 10**

Let $C^1(a, b)$ be the set of functions continuous on $(a, b)$, and suppose $S$ is the subset of $C^1(a, b)$ that consists of the set of differentiable functions on $(a, b)$. Is $S$ a subspace of $C^1(a, b)$?

**Solution**

By definition, $S$ is a subset of $C^1(a, b)$. Since $z(x) = 0$ is differentiable ($z'(x) = 0$), the zero vector is in $S$. Also, from calculus we know that sums and constant multiples of differentiable functions are differentiable, so that $S$ is closed under addition and scalar multiplication. Hence $S$ is a subspace of $C^1(a, b)$.

In this chapter the proofs of many theorems are left as exercises because their Euclidean counterparts have very similar proofs given earlier in the book.

**Example 11**

Let $C^1(\mathbb{R})$ denote the set of functions that are continuous on all of $\mathbb{R}$. Let $S$ denote the subset of $C^1(\mathbb{R})$ consisting of functions $y(t)$ that satisfy the differential equation

$$y''(t) + y(t) = 0 \quad (1)$$

Show that $S$ is a subspace of $C^1(\mathbb{R})$.

**Solution**

Differential equations of this type arise in the modeling of simple harmonic motion, such as a mass moving up and down while suspended by a spring (see Figure 1).

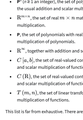



Figure 1 A mass-spring system. $y(t)$ gives the vertical displacement at time *t*.

To show that $S$ is a subspace, we start by noting that if $y(t) = 0$ is the zero function, then $y''(t) = 0$ and this function satisfies (1).

Next, suppose that $y_1$ and $y_2$ both satisfy (1). Then

$$\begin{aligned}(y_1(t) + y_2(t))'' + (y_1(t) + y_2(t)) &= y_1''(t) + y_2''(t) + y_1(t) + y_2(t) \\ &= (y_1''(t) + y_1(t)) + (y_2''(t) + y_2(t)) = 0\end{aligned}$$

so $y_1 + y_2$ is also in $S$.

Finally, given a scalar $c$ and a solution $y_1$ of (1), we have

$$(cy_1(t))'' + cy_1(t) = c(y_1''(t) + y_1(t)) = 0$$

Thus $cy_1$ is also in $S$, and therefore $S$ is a subspace.

We close this lesson with a reminder. As mentioned earlier, often it is difficult to transition away from the notion that elements of $\mathbf{R}^n$ are the only type of "vector." As we have seen, viewed from a more general perspective, we can think of polynomials, matrices, and continuous functions as vectors. So remember: Vectors are not always columns of numbers!

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